

Problem Set 3 - Linear Algebra ¹

Due date: 09/08

1. Consider a random sample $\mathbf{X}_1, \dots, \mathbf{X}_n$ from a distribution with a mean μ and variance σ^2 . Consider the variance estimator $\hat{\sigma}_n^2$ defined as:

$$\hat{\sigma}_n^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X}_n)^2$$

where $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$ is the sample mean.

- (a) Show that $\hat{\sigma}_n^2$ is a biased estimator of σ^2 .
 - (b) What is the bias?
 - (c) How can you modify $\hat{\sigma}_n^2$ to make it unbiased?
2. Let $X_1, \dots, X_n$ be i.i.d. random variables with variance σ^2 . Consider the two estimators of σ^2 :

$$\hat{\sigma}_n^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X}_n)^2 \quad \text{and} \quad S_n^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X}_n)^2$$

Recall from the lecture notes that:

$$\begin{aligned} \text{Var}(S_n^2) &= \frac{2}{n-1} \sigma^4 & \text{and} & & \text{Var}(\hat{\sigma}_n^2) &= \frac{2(n-1)}{n^2} \sigma^4 \\ \text{MSE}(S_n^2) &= \frac{2}{n-1} \sigma^4 & \text{and} & & \text{MSE}(\hat{\sigma}_n^2) &= \frac{2n-1}{n^2} \sigma^4 \end{aligned}$$

The *MSE* of an estimator $\hat{\theta}$ for a parameter θ can be decomposed as:

$$MSE(\hat{\theta}) = \text{Var}(\hat{\theta}) + \text{Bias}(\hat{\theta})^2$$

- (a) Find the expression for the ratio $\frac{\text{Var}(S_n^2)}{\text{Var}(\hat{\sigma}_n^2)}$.
- (b) Find the expression for the ratio $\frac{\text{MSE}(S_n^2)}{\text{MSE}(\hat{\sigma}_n^2)}$.
- (c) Do a ggplot graph in R to visualize the two ratios above as a function of n (start your graphs with $n > 30$). What do you observe? (Report the graphs and code)

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3. Consider the simple linear regression model:

$$Y_i = \beta_0 + \beta_1 X_i + u_i, \quad i = 1, \dots, n,$$

where $\{(X_i, Y_i)\}_{i=1}^n$ is an i.i.d. sample. Assume $\mathbb{E}[u_i] = 0$ and $\mathbb{E}[u_i^2] = \sigma^2 < \infty$.

Method of Moments Estimator

(a) Using the moment conditions

$$\mathbb{E}[u_i] = 0, \quad \mathbb{E}[X_i u_i] = 0,$$

derive the Method of Moments estimator $(\hat{\beta}_0^{MM}, \hat{\beta}_1^{MM})$.

(b) Show that the Method of Moments estimator coincides with the OLS estimator in the simple regression.

Maximum Likelihood Estimator

Assume now that the sampling distribution satisfies:

$$u_i \sim \mathcal{N}(0, \sigma^2) \quad \text{independently across } i.$$

(c) Derive the MLEs $\hat{\beta}_0^{MLE}$, $\hat{\beta}_1^{MLE}$, and $\hat{\sigma}_{MLE}^2$.

(d) Show that the MLEs for β_0 and β_1 coincide with the OLS estimator.

Consistency of $\hat{\beta}_1$

(e) Show that $\hat{\beta}_1$ is a consistent estimator of β_1 . (Hint: Express $\hat{\beta}_1$ as a ratio of sample averages and apply the Law of Large Numbers.)

Asymptotic Distribution of $\hat{\beta}_1$

(f) Derive the asymptotic distribution of $\hat{\beta}_1$. (Hint: Use the Central Limit Theorem and Slutsky's Theorem.)

4. Randomization Inference

Consider the following extremely small sample from the [GAIN Experiment](#).

Table 1: Six Observations from the GAIN Experiment

Individual	Potential Outcome $Y_i(0)$	Potential Outcome $Y_i(1)$	Treatment D_i	Observed Outcome Y_i^{Obs}
1	66	?	0	66
2	0	?	0	0
3	0	?	0	0
4	?	0	1	0
5	?	607	1	607
6	?	436	1	436

The fundamental problem of causal inference is that we can never observe both potential outcomes for any single individual. We only observe:

$$Y_i^{Obs} = D_i \cdot Y_i(1) + (1 - D_i) \cdot Y_i(0)$$

A common estimator for the Average Treatment Effect (ATE) is the simple difference in means from the sample:

$$\hat{\tau} = \frac{1}{N_1} \sum_{i:D_i=1} Y_i - \frac{1}{N_0} \sum_{i:D_i=0} Y_i$$

where N_1 and N_0 are the number of treated and control units, respectively.

Use R to solve the questions below and report your code.

Calculating $\hat{\tau}$

- Calculate the mean outcome for the treatment group ($D_i = 1$).
- Calculate the mean outcome for the control group ($D_i = 0$).
- Calculate the simple difference in means ($\hat{\tau}$). Based **only** on this result, what would you conclude about the effect of the program?

Fisher's Exact Test

We will now use Randomization Inference to test the sharp null hypothesis of no treatment effect for any individual, formalized as:

$$H_0 : Y_i(1) = Y_i(0) \quad \forall i$$

Under this null hypothesis, all potential outcomes are known and fixed. The observed differences are solely due to the random assignment of treatment D_i . Notice that if we assume the null is true, table 1 becomes:

Table 2: Six Observations from the GAIN Experiment (Under Null Hypothesis)

Individual	Potential Outcome $Y_i(0)$	Potential Outcome $Y_i(1)$	Treatment D_i	Observed Outcome Y_i^{Obs}
1	66	66	0	66
2	0	0	0	0
3	0	0	0	0
4	0	0	1	0
5	607	607	1	607
6	436	436	1	436

And we can derive the **randomization distribution** of any test statistic: a function of the assignment vector $\mathbf{D}$ and the observed outcomes $\mathbf{Y}^{obs}$. This statistic varies only because of the random assignment of treatment ($\mathbf{D}$), not because of any uncertainty about the potential outcomes (there is no missing data issue).

Consider the test statistic $T(\mathbf{D}, \mathbf{Y}^{obs})$, basically $\hat{\tau}$:

$$T = T(\mathbf{D}, \mathbf{Y}^{obs}) = \frac{1}{3} \sum_{i=1}^6 D_i \cdot Y_i^{obs} - \frac{1}{3} \sum_{i=1}^6 (1 - D_i) \cdot Y_i^{obs}.$$

- (a) For the observed assignment vector $\mathbf{D}^{obs} = (0, 0, 0, 1, 1, 1)$, calculate the value of the test statistic, T . You can call this T^{obs} (Repeating (c) from last question.)
- (b) How many possible $\mathbf{D}$ vectors are there for assigning the three treatment assignments ($D_i = 1$) to these six individuals? This is the number of possible permutations of the assignment vector.
- (c) The **randomization distribution** of the test statistic is generated by calculating $T(\mathbf{D}, \mathbf{Y}^{obs})$ for **every possible assignment vector** $\mathbf{D}$, holding the vector of outcomes fixed. For each possible assignment, calculate the T .
- (d) Plot a histogram of T calculated under all possible random assignments. This histogram represents the exact distribution of the test statistic under the sharp null hypothesis.
- (e) On the histogram, mark the value of T^{obs} calculated in (a).
- (f) Find the 2.5th percentile and the 97.5th percentile of this randomization distribution. These form a 95% reference interval for the test statistic under the null hypothesis.
- (g) Is the observed test statistic T^{obs} from part (a) inside or outside this 95% interval? Based on this, would you reject or fail to reject the null hypothesis H_0 at the 5% significance level?
- (h) **(Optional)** Calculate the two-sided **p-value**. This is the probability, under the null hypothesis, of observing a value of the test statistic that is as extreme in absolute value than the one actually observed. Formally, it is the proportion of assignment vectors for which $|T(\mathbf{D}, \mathbf{Y}^{obs})| \geq |T^{obs}|$.
- (i) **(Optional)** Based on this p-value, what do you conclude about the null hypothesis?

For larger datasets, the Fisher exact test becomes computationally intensive and is therefore rarely used in practice. However, [this paper](#) applies randomization inference to re-evaluate several high-profile RCTs and illustrates how the framework can be used in modern empirical work.